

MILESTONE 1

SW Engineering CSC648-848 Spring 2025

Buy-Sell WWW app - GatorMarket

Team 2

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1. **Executive Summary**

San Francisco State University is home to a diverse community of students, faculty, and staff with unique needs for buying and selling goods. Whether it's textbooks, electronics, or furniture, members of the SFSU community often seek affordable items or ways to declutter their belongings. Our product, GatorMarket, aims to bridge this gap by providing an online marketplace exclusively for the SFSU community customized to the needs of this community. This platform will empower users to buy, sell, and coordinate on-campus pickups, fostering both sustainability and community engagement.

GatorMarket will offer an intuitive and user-friendly interface where users can create accounts using their SFSU email, ensuring that only verified members of the community participate. Created by the very students that will be using this product we will be able to implement game changing features such as 'class search' allowing for students to find and purchase class materials specific for their course load. Furthermore the exclusiveness of this product will ensure safety for our users as well as convenience. Products come to you as you can schedule your pick up at designated safe pick up locations on campus. Unlike competitors such as Craigslist or Facebook, GatorMarkets's design is kept simple, with no need for complicated chat features or public postings. This format gives the user exactly what they are looking for, a platform to post and purchase the products they need.

Our team consists of five dedicated computer science majors who understand the unique needs of the SFSU community. With firsthand experience navigating campus life, we are committed to building an accessible, secure, and user-friendly platform. We believe that GatorMarket will become an essential tool for enhancing campus life and fostering a stronger sense of community at SFSU.

2. Personae



1. SFSU Student Buyer - Alex

Alex is a second-year SFSU student with strong online research skills, whether it's tracking down class resources or figuring out the fastest way to get to a new building. He balances a busy class schedule with a tight budget and often needs textbooks and course materials for multiple classes. Alex is resourceful and comfortable navigating digital platforms to find what he needs.

Goals:

- Find cheap textbooks and electronics, wants an easy way to find the books he'll need for his specific classes
- He'd like to not have to go out of his way to pick up his purchases.
- Some online buy sells can be sketchy, Alexs wants to confirm the buyers are safe and reliable

Skills:

- Lots of experience online and using other similar applications (ex. Facebook Marketplace)
- Knows how to use key words and advanced search features

Pain Points:

- Has to have the materials he needs for class on Monday and doesn't have time to deal with unreliable sellers.
- Struggles to find specific textbooks without clear course-based filters.



2. SFSU Student Seller - Maria

Maria is a graduating senior at SFSU who is focused on preparing for her post-grad move. She is organized and efficient, often managing large tasks like packing and coordinating her schedule. With several personal belongings accumulated over her college years, Maria is motivated to declutter and stay productive.

Goals:

- Sell her old furniture, clothes, and books quickly
- Wants to help the students that will need these books, she remembers how hard they were to find.

Skills:

- Has experience selling through online platforms and understands pricing and needed information about products.
- Great tech skills, familiar with similar sites.

Pain Points:

- Doesn't have a lot of time in her schedule and buyers on other sites are asking her to ship the goods. This will take longer and be costly.



3. SFSU Admin - Lisa

Lisa is a GatorMarket staff member who supports the site day to day activities. She is detail-oriented and committed to ensuring that campus spaces, both physical and digital, are welcoming, safe, and compliant with community standards. Lisa regularly oversees processes that require reviewing submissions and enforcing guidelines. She reviews new posts and monitors for inappropriate content before the post is made public.

Goals:

- Review and approve new listings before they go live on GatorMarket.
- Keep the platform organized and safe by enforcing SFSU community standards.

Skills:

- Experienced in content moderation through the admin review tools.
- Knowledgeable of the community and what is considered appropriate

Pain Points:

- Managing a high volume of new listings in the admin queue.



4. New SFSU Student - Jordan

Jordan is a first-year SFSU student living on campus. He is still adjusting to college life and is new to online platforms for transactions. Jordan values simplicity and direct communication and prefers easy, straightforward ways to handle everyday needs.

Goals:

- Buy affordable dorm essentials
- Coordinate pickups with sellers nearby because he lives on campus

Skills:

- Tech knowledge good with search filters and product verification

Pain Points:

- Unfamiliar with campus and needs to find easy spots to meet sellers
- Needs a lot of dorm essentials and does not have a big budget
- Still new to the community so he doesn't know many people that can help him out

3. High-level Use cases

1. Searching and Browsing Listings

Alex is an SFSU student looking for a used textbook, he visits GatorMarket. He is not sure if the book is available, so he browses through the textbooks and course material category and uses the **search** function to **filter** results by title and course name. He browses the items for sale and compares the images and prices before deciding to contact a seller. After typing the **message** and clicking send Alex is prompted to **login** or **register** using his SFSU email.

2. Posting an Item for Sale

Maria is a graduating senior and wants to sell her old furniture before moving out. She goes to GatorMarket, navigates to the **post** item section, provides details about the furniture, and is prompted to **login** in using her SFSU email when clicking post. She submits her listing and is informed it may take up to 24 hours before it is approved. Once approved, her **post** is visible to other students, and she can wait to be **messaged** by potential buyers.

3. Contacting a Seller

Jordan finds a used laptop for sale and wants to negotiate the price. He clicks on the listing then the **message** button in order to contact the seller through the site's messaging system. The seller receives the **message** in their dashboard. Jordan checks back later for the seller's response and arranges to meet for the transaction. Using the map on GatorMarket that provides safe locations to meet on campus, Jordan finds somewhere to meet the seller. After confirming the **location**, Jordan and the seller meet and successfully complete the transaction.

4. Admin Review

Lisa is an **admin** for GatorMarket and logs in to review new listings before approving them. She checks for inappropriate content, duplicate listings, or policy violations. When she finds a listing selling an unauthorized service, she denies it and notifies the user.

4. List of main data items and entities

1. User types and permissions

1.1 Registered User:

- **Definition:** SFSU students and faculty have access to the system.

- **Permissions:**
 - Credentials
 - Access to Website
 - View Posts
 - Send and receive messages
 - Post items

1.2 Unregistered User:

- **Definition:** anyone can see the website and interact with items but cannot message or post
- **Permissions:**
 - Create Credentials (become a registered user)
 - Access to Website
 - View Posts

1.3 Admin:

- **Definition:** User to oversee all posts.
- **Permissions:**
 - All registered users permissions
 - Manage the posts and check for appropriateness, approve pending posts

2. Core Entities

2.1 User sessions:

- **Definition:** Active user in session
- **Usage:**
 - Tracks user authentication
 - Stores user's session data temporally
 - Contains **Active Users**

2.2 Message:

- **Definition:** A unit of communication within the chat room
- **Usage:**
 - Contains text sent by users
 - Is associated with the sender
 - Can be stored in a chat history
 - Can include a timestamp of when the message was received and sent

2.3 User Credentials:

- **Definition:** Authentication details for system access.
- **Usage:**

- SFSU students and faculty Email and password.
- Used to validate login into the server

2.4 Active Users:

- **Definition:** displays how many users are on the server.
- **Usage:**
 - Displayed in the UI to show participants.
 - Updated dynamically as users join or leave.

2.5 Filters

- **Definition:** users can search for items
- **Usage:**
 - Enables users to search for items
 - Supports filters and categories for refined searches.

2.6 Product Listings

- **Definition:** Items posted by users for sale or exchange.
- **Usage:**
 - Allows users to post product details, descriptions, and prices.
 - Includes categories for better organization.
 - Can be searched and filtered via the **Filters**.

3. SFSU custom entities

3.1 School Locations:

- **Definition:** locations where you can meet up or drop off
- **Usage:**
 - Helps associate chat rooms with specific classrooms, lecture halls, or study areas.
 - Can be used for location-based filtering of chat groups (e.g., chat rooms for students in a specific building).
 - Supports event notifications or campus-specific announcements.

5. List of high level functional requirements

Unregistered Users (Public Access)

1. Users shall be able to browse all available listings without an account.
2. Users shall be able to search for items by title, category, or keywords.
3. Users shall be able to search items by category and text input.
4. Users shall be able to filter items by price.
5. Users shall be able to view item descriptions, images and item status (available, pending, sold).
6. Users shall be able to register.
7. Students shall be able to search for books based on specific SFSU courses (e.g., CSC 648-848) to find textbooks for their classes.

Registered Users (SFSU Students Only)

8. Only members of the SFSU community shall be allowed to sign up and log in using their university email.
9. Users shall be able to create a new listing with a title, description, price, category, and images and post it (subject to admin approval).
10. Users shall be able to remove their active listings from their dashboard.
11. Users shall be able to message the seller.
12. Sellers shall have a dashboard showing all their active listings and messages.
13. Sellers shall be able to delete an item once a transaction is complete.
14. Users shall be able to login.

Admin (Moderators)

15. Admin shall be required to review and approve an item before becoming public.
16. Admins shall have the ability to delete listings that violate platform policies.
17. Admins shall have access to an overview of all listings, including reported ones.

Special SFSU-Specific Features

18. Users shall be able to specify a meet up place (e.g., library, student center) for easy trading.

6. **List of non-functional requirements**

1. Application shall be developed, tested and deployed using tools and cloud servers approved by Class CTO and as agreed in M0

2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All or selected application functions shall render well on mobile devices (no native app to be developed)
4. Posting of sales information and messaging to sellers shall be limited only to SFSU students
5. Critical data shall be stored in the database on the team's deployment server.
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application shall follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used
13. No email clients or chat services shall be allowed. Interested users can only message to sellers via in-site messaging. One round of messaging (from user to seller) is enough for this application
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Media formats shall be standard as used in the market today
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "SFSU Software Engineering Project CSC 648-848, Spring 2025. For Demonstration Only" at the top of the WWW page Nav bar. (Important so as to not confuse this with a real application). You have to use this exact text without any editing.

7. Competitive Analysis

Features	GatorMarket	Facebook Marketplace	Rumie	Craigslist
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Search	+	+	+	+
Messaging	+	+	+	-
Categories	+	+	+	+
SFSU class specifics	+	-	-	-
SFSU Meetup locations	+	-	-	-
Student Verification	+	-	-	-

+ exists, - does not exist

GatorMarket, like most other marketplace websites will have search, messaging, and item categories but what makes it unique is that it is made for SFSU students. Students will be able to search specifically for materials, and textbooks based on classes at SFSU, making it much easier to find items that they are looking for. Additionally, GatorMarket prioritizes student safety by introducing designated safe trade locations on and around the SFSU campus, which competitors do not provide. To further enhance security, student verification ensures that all users are actual SFSU students, reducing the risk of scams and making transactions trustworthy.

While Facebook Marketplace and Craigslist serve a broad audience, and Rumie focuses on educational materials, GatorMarket is the only platform designed to meet the specific needs of SFSU students, making it a more relevant, secure, and efficient solution for campus-based transactions.

8. High-level system architecture and technologies used

1. Main Software Components and Versions

- **Server Host:** AWS EC2
- **Operating System:** Ubuntu 24.04 Server
- **Database:** MySQL v8.0
- **Web Server:** NGINX 1.26
- **Server-Side Language:** JavaScript (Node.js)

- **Web Framework:** Express.js

2. Deployment Cloud Service

- AWS EC2 for hosting
- **Cert Bot (Let's Encrypt)** for SSL certificate management

3. Frontend Frameworks

- **React.js 19**
- **SASS 1.85.0** for styling

4. Supported Browsers

- **Google Chrome** (latest two versions)
- **Safari** (latest two versions for macOS & iOS)

5. External Open-Source APIs and Tools

Google Analytics for tracking and monitoring user interactions

9. Use of GenAI tools like ChatGPT and copilot for Milestone 1

We decided to use ChatGPT to assist in creating our Milestone 1 document. The primary role of ChatGPT was creating a mock up or outline that we then added information specific to our project. Google Gen AI was also used for research purposes.

For the Summary, Use Cases, Functional Requirements, and Data Entries ChatGPT was used to create a rough draft. Simply inputting the section description and asking ChatGPT to generate a response we obtained an outline that served as inspiration for our writing.

Helpfulness: HIGH

Example Prompt

“I need high-level use cases for a website where SFSU students will be able to buy and sell products. Use cases should specify a specific scenario in which someone is using the site.”

The persona images were created through chat gpt given the description of the person and their needs. Without AI it would have been a lot harder to find appropriate profile photos that represented our personae and were not protected intellectual property.

Helpfulness: HIGH

Example Prompt

“Given the following persona description generate an image of a person that fits this description”

The website competitors were found through a google search within which google AI gave us a list of likely competitors for our site. While this did give us great information, in this situation it would not have been very hard to find these results without the AI.

Helpfulness: MEDIUM

Example Search

“campus specific buy sell sites”

Output

“For buying and selling items on campus, consider platforms like Vezzy, rumie, CampusStore, Campus Street, and TreasureX, which are designed for college students to connect and trade within their specific campus communities.”

Some team members also attempted to use ChatGPT to assist them in their peer review. In my opinion this was less than helpful as AI might not consider the specific requirements of this assignment and does not give the reviewer the opportunity to deepen their understanding of the project mission.

Helpfulness: LOW

Example Prompt

“Give feedback on this writeup”

10. Team and roles

Name	Email	Role
Claudia Wormley	cwormley@sfsu.edu	Team Leader
Nathan Donat-Filliod	ndonatfilliod@sfsu.edu	Github Master
Daniel Cervantes	dcervantes2@sfsu.edu	Back-End Team Lead
Davis Rosenstein	drosenstein@sfsu.edu	Front-End Team Lead
Fatimah Abdolcader	fabdolcader@sfsu.edu	Front-End Team Member

11. Team Lead Checklist to be completed by team lead

- a. So far all team members are fully engaged and attending team sessions when required **DONE**
- b. Team found a time slot to meet outside of the class **ON TRACK**
- c. Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing **DONE**
- d. Team reviewed class slides on requirements and use cases before drafting Milestone 1 **DONE**
- e. Team reviewed non-functional requirements from “How to start...” document and developed Milestone 1 consistently **DONE**
- f. Team lead checked Milestone 1 document for quality, completeness, formatting and compliance with instructions before the submission **DONE**
- g. Team lead ensured that all team members read the final M1 and agree/understand it before submission **DONE**
- h. Team shared and discussed experience with GenAI tools among themselves **DONE**
- i. Github organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.) **DONE**